

🎮 Level-Up Mathematics: Applied Probability 🏆

1. ☁️ Level 1: Cloud Gaming Chaos

A cloud gaming provider, like Nvidia G-Force Now, hosts dedicated streaming servers 🖥️ for an esports tournament 🏆 at the beginning of each month. Assume that the number of connection drops 📶 for a server in a month follows the Poisson distribution with mean 1.9. Suppose there are totally 15 servers allocated for the tournament, and the performance of a server in a month is regarded as unacceptable 🚫 if there are more than 2 connection drops in that month. Assume that the monthly numbers of connection drops of the servers are independent.

- (a) (2 points) Find the probability that the performance of a randomly selected server 🖥️ in a certain month is unacceptable.

Solution: Let X be the number of connection drops in a month. $X \sim \text{Po}(1.9)$.
We want $P(X > 2) = 1 - P(X \leq 2)$:

$$\begin{aligned} P(X > 2) &= 1 - e^{-1.9} \left(\frac{1.9^0}{0!} + \frac{1.9^1}{1!} + \frac{1.9^2}{2!} \right) \\ &= 1 - e^{-1.9}(1 + 1.9 + 1.805) \\ &\approx 1 - 0.7037 = 0.2963 \end{aligned}$$

R Code:

```
1 # lower.tail = FALSE calculates P(X > x)
2 p_unacceptable <- ppois(2, lambda = 1.9, lower.tail = FALSE)
3 print(p_unacceptable)
```

- (b) (2 points) For a certain server, find the probability that June of 2026 is the 3rd month in 2026 such that the performance of that server is unacceptable.

Solution: For June (the 6th month) to be the 3rd unacceptable month, two things must happen independently: 1. There must be exactly 2 unacceptable months in the first 5 months (Jan-May). 2. June itself must be an unacceptable

month. Let $p = 0.2963$ be the probability of an unacceptable month.

$$\begin{aligned} P &= \left[\binom{5}{2} p^2 (1-p)^3 \right] \times p \\ &= [10 \times (0.2963)^2 \times (0.7037)^3] \times 0.2963 \\ &\approx 0.3056 \times 0.2963 \approx 0.0906 \end{aligned}$$

R Code:

```
1 # Binomial prob of 2 in 5, multiplied by prob of success on 6th
2 p_first_5 <- dbinom(2, size = 5, prob = p_unacceptable)
3 p_june <- p_first_5 * p_unacceptable
4 print(p_june)
```



Teacher's Note

Since we cannot use the Negative Binomial distribution directly, break it down logically for the students: "Find the Binomial probability for the preceding trials, then multiply by the final required outcome."

- (c) (2 points) Find the expected total number of unacceptable  server performances for all 15 servers over one year.

Solution: Total server-months = 15 servers $\times$ 12 months = 180 trials.

Expected value $E(Y) = n \times p$:

$$E(Y) = 180 \times 0.2963 = 53.334$$

R Code:

```
1 expected_value <- 15 * 12 * p_unacceptable
2 print(expected_value)
```

- (d) (4 points) In order to assure the stream quality , the tournament organizers introduce the following term in the new hosting contract for the 15 servers. For each server , if its performance is unacceptable for 3 consecutive months, one hardware replacement order  will be immediately issued. Find the probability that 3 or more hardware replacement orders  will be issued to the provider on or before 30th April 2027.

Solution: First, find the probability one server gets replaced within the 4 months (Jan-Apr).

Replaced if unacceptable (U) for 3 consecutive months. Acceptable is (A).

Possible winning sequences: 1. U-U-U in Jan, Feb, Mar: $P = p^3$ 2. A-U-U-U in Jan, Feb, Mar, Apr: $P = (1 - p)p^3$

$$p_{rep} = p^3 + (1 - p)p^3 = p^3(2 - p) \approx 0.0443.$$

Now, let Y be the number of servers replaced. $Y \sim \text{Bin}(15, 0.0443)$.

$$P(Y \geq 3) = 1 - P(Y \leq 2) = 1 - 0.9730 = 0.0270$$
 R Code:

```
1 p <- p_unacceptable
2 p_rep <- p^3 * (2 - p)
3 # P(Y >= 3) is 1 - P(Y <= 2)
4 p_3_or_more <- pbinom(2, size = 15, prob = p_rep, lower.tail = FALSE)
5 print(p_3_or_more)
```

2. 🎮 Level 2: The VR Lag Spikes

The number of lag spikes ⚡ in a day while playing on a Beta Quest 3 VR headset 🎮 follows the Poisson distribution with mean 4.8.

(a) (2 points) Find the probability that there are not more than 3 lag spikes ⚡ in a day.

Solution: Let $L \sim \text{Po}(4.8)$. We want $P(L \leq 3)$:

$$P(L \leq 3) = e^{-4.8} \left(1 + 4.8 + \frac{4.8^2}{2} + \frac{4.8^3}{6} \right) \approx 0.2942$$

R Code:

```
1 p_le_3 <- ppois(3, lambda = 4.8)
2 print(p_le_3)
```

(b) (2 points) Find the probability that, in 3 consecutive days, there are at most 2 days with not more than 3 lag spikes in each day.

Solution: This is the complement of having ≤ 3 lag spikes on all 3 days.

$$P(\text{at most 2 days}) = 1 - P(\text{all 3 days}) = 1 - (0.2942)^3 \approx 0.9745.$$
 R Code:

```
1 p_at_most_2 <- pbinom(2, size = 3, prob = p_le_3)
2 print(p_at_most_2)
```

- (c) A day is called a rage day 😡 if there are more than 5 lag spikes in that day; otherwise it is called a smooth day 😊.
- i. (2 points) Suppose today is a rage day 😡. Find the mean number of smooth days 😊 between today and the next rage day.

Solution: First, find the probabilities of Smooth (S) and Rage (R) days:

$$P(S) = P(L \leq 5) \approx 0.6510$$

$$P(R) = 1 - 0.6510 = 0.3490$$

Let E be the expected number of smooth days before the next rage day. We can set up an algebraic equation based on what happens tomorrow:

- If tomorrow is a rage day (prob $P(R)$), we add 0 smooth days and stop.
- If tomorrow is a smooth day (prob $P(S)$), we add 1 smooth day, and the expected number of remaining smooth days is still E .

$$E = P(R) \cdot 0 + P(S) \cdot (1 + E)$$

$$E = P(S) + P(S) \cdot E$$

$$E - P(S) \cdot E = P(S)$$

$$E(1 - P(S)) = P(S)$$

$$E(P(R)) = P(S)$$

$$E = \frac{P(S)}{P(R)} = \frac{0.6510}{0.3490} \approx 1.865$$

R Code:

```
1 p_smooth <- ppois(5, lambda = 4.8)
2 p_rage <- 1 - p_smooth
3 mean_smooth <- p_smooth / p_rage
4 print(mean_smooth)
```



Teacher's Note

This recursive expectation method is a brilliant algebraic way to find expected waiting times without ever teaching the Geometric distribution formula.

- ii. (2 points) Find the probability that the last day of a week is the third rage day

in that week.

Solution: For the 7th day to be the 3rd rage day, two events must happen:
1. There must be exactly 2 rage days in the first 6 days (Binomial). 2. The 7th day must be a rage day.

$$\begin{aligned} P &= \left[\binom{6}{2} P(R)^2 P(S)^4 \right] \times P(R) \\ &= [15 \times (0.3490)^2 (0.6510)^4] \times 0.3490 \\ &\approx 0.3282 \times 0.3490 \approx 0.1145 \end{aligned}$$

R Code:

```
1 # Binomial prob for the first 6 days, then success on 7th
2 p_first_6 <- dbinom(2, size = 6, prob = p_rage)
3 p_7th_day <- p_first_6 * p_rage
4 print(p_7th_day)
```

- iii. (2 points) Find the probability that there are at least 4 consecutive rage days 🙄 in a week.

Solution: Let $p = 0.3490$ (Rage) and $q = 0.6510$ (Smooth). In 7 days, a run of ≥ 4 can start on day 1, 2, 3, or 4. Total Probability $= p^4 + 3qp^4 = p^4(1 + 3q)$:

$$P = (0.3490)^4(1 + 3(0.6510)) \approx 0.0437$$

R Code:

```
1 p_streak <- (p_rage^4) * (1 + 3 * p_smooth)
2 print(p_streak)
```

3. 💎 Level 3: Gacha Grinding

Wishes 📦 per minute follows Poisson($\lambda = 3.2$). The drop rate for “Masterless Star-glitters” ★ per Wish is given.

- (a) (3 points) Find the probability that fewer than 4 Wishes 📦 are opened by a player in a certain minute.

Solution: Let $W \sim \text{Po}(3.2)$. We want $P(W < 4) = P(W \leq 3)$.

$$P(W \leq 3) = e^{-3.2}(1 + 3.2 + 5.12 + 5.461) \approx 0.6025$$

R Code:

```
1 p_wishes_lt_4 <- ppois(3, lambda = 3.2)
2 print(p_wishes_lt_4)
```

- (b) (2 points) Find the probability that the 8th Wish opened is the 3rd Wish that drops exactly 2 Masterless Star-glitters ★.

Solution: Probability of exactly 2 glitters is $p = 0.7$. For the 8th wish to be the 3rd success: 1. Exactly 2 successes in the first 7 wishes (Binomial). 2. The 8th wish is a success.

$$\begin{aligned} P &= \left[\binom{7}{2} (0.7)^2 (0.3)^5 \right] \times 0.7 \\ &= [21 \times 0.49 \times 0.00243] \times 0.7 \\ &\approx 0.02500 \times 0.7 \approx 0.0175 \end{aligned}$$

R Code:

```
1 # Binomial prob for the first 7 wishes, then success on 8th
2 p_first_7 <- dbinom(2, size = 7, prob = 0.7)
3 p_8th_wish <- p_first_7 * 0.7
4 print(p_8th_wish)
```

- (c) (2 points) Find the probability that exactly 3 Wishes are opened in a certain minute and each of them drops exactly 2 Masterless Star-glitters.

Solution: Intersection of two independent events: $P(W = 3) \times P(\text{all 3 drop 2})$.

$$\text{Total } P = \left(e^{-3.2} \frac{3.2^3}{6} \right) \times (0.7)^3 \approx 0.2226 \times 0.343 \approx 0.0764$$

R Code:

```
1 p_3_wishes_all_2 <- dpois(3, lambda = 3.2) * (0.7^3)
2 print(p_3_wishes_all_2)
```

- (d) (3 points) Find the probability that exactly 3 Wishes 📦 are opened in a certain

minute and they drop a total of 6 Masterless Star-glitters ★.

Solution: Ways for 3 wishes to sum to 6 glitters:

- 2, 2, 2: $(0.7)^3 = 0.343$
- 1, 2, 3 (6 perms): $6 \times (0.12 \times 0.7 \times 0.08) = 0.04032$
- 1, 1, 4 (3 perms): $3 \times (0.12 \times 0.12 \times 0.04) = 0.001728$

Final Probability = $P(W = 3) \times 0.38505 \approx 0.2226 \times 0.38505 \approx 0.0857$ R Code:

```
1 p_sum_6 <- (0.7^3) + 6*(0.12 * 0.7 * 0.08) + 3*(0.12 * 0.12 * 0.04)
2 p_final <- dpois(3, lambda = 3.2) * p_sum_6
3 print(p_final)
```

4. 🎮 Level 4: The Live Event Login Queue

Tom joins the login queue ⌚ for a massive live event in Fort-night.

- (a) (2 points) Find the probability that Tom successfully bypasses the queue ⌚ on or before 7:30 PM on a certain day.

Solution: He gets in at 7:20, OR fails 7:20 and gets in at 7:30.

$$P(\text{Bypass}) = 0.9 + (0.1 \times 0.9) = 0.99$$

R Code:

```
1 p_bypass <- 0.9 + (0.1 * 0.9)
2 print(p_bypass)
```

- (b) (2 points) Find the probability that Tom is late to the event 🕒 on a certain day.

Solution: Law of Total Probability over 3 branches:

$$\begin{aligned} P(\text{Late}) &= (0.9 \times 0.1) + (0.1 \times 0.9 \times 0.4) + (0.1 \times 0.1) \\ &= 0.09 + 0.036 + 0.01 = 0.136 \end{aligned}$$

R Code:

```
1 p_late <- (0.9 * 0.1) + (0.1 * 0.9 * 0.4) + (0.1 * 0.1)
2 print(p_late)
```

- (c) (2 points) Find the probability that Tom is late to the event 2 times out of 6 event days.

Solution: Binomial with $n = 6, p = 0.136$.

$$P(\text{Late twice}) = \binom{6}{2}(0.136)^2(0.864)^4 \approx 0.1546$$

R Code:

```
1 p_late_twice <- dbinom(2, size = 6, prob = p_late)
2 print(p_late_twice)
```

- (d) 7 players in a Discord voice channel 🎧. If Tom is late, he plays Casual Mode (others: 0.7 Casual, 0.3 Ranked); otherwise he plays Ranked League.

- i. (2 points) Find the probability that the 7 players are playing the same game mode.

Solution: Scenario A (All Casual): $0.136 \times 0.7^6 \approx 0.0160$

Scenario B (All Ranked): $0.864 \times 0.3^6 \approx 0.0006$

Total $P = 0.0160 + 0.0006 = 0.0166$ R Code:

```
1 p_all_casual <- p_late * (0.7^6)
2 p_all_ranked <- (1 - p_late) * (0.3^6)
3 p_same_mode <- p_all_casual + p_all_ranked
4 print(p_same_mode)
```

- ii. (2 points) Find the probability that exactly 3 players are playing Ranked League 🏆.

Solution: Scenario A: Tom is Ranked (0.864). We need exactly 2 of the 6 others to be Ranked. $P_A = 0.864 \times \binom{6}{2}(0.3)^2(0.7)^4 = 0.2800$

Scenario B: Tom is Casual (0.136). We need exactly 3 of the 6 others to be Ranked. $P_B = 0.136 \times \binom{6}{3}(0.3)^3(0.7)^3 = 0.0252$

Total $P = 0.2800 + 0.0252 = 0.3052$ R Code:

```
1 # dbinom calculates the exact probability for the "other 6" players
2 p_tom_ranked_2_others <- (1 - p_late) * dbinom(2, size = 6, prob = 0.3)
3 p_tom_casual_3_others <- p_late * dbinom(3, size = 6, prob = 0.3)
4 p_exactly_3_ranked <- p_tom_ranked_2_others + p_tom_casual_3_others
5 print(p_exactly_3_ranked)
```